Competition and Dependence: Mexico, China, and the New Export Routes to the United States

Vladimir Márquez Stone* Seyka Sandoval**

Abstract

This article analyzes the recent reconfiguration of Global Value Chains (GVC) among Mexico, China, and the United States in the context of the USMCA, nearshoring, and the global commercial tensions associated with the second China shock. Using bilateral trade data at the product level (HS6) from UN Comtrade from January 2016 to November 2024, we identify changes in direct competition between Mexico and China in the US market, as well as Mexico's persistent reliance on Chinese intermediate inputs. Using a difference-in-differences design with product and year-month fixed effects, we estimate that sectors where China held a higher pre-tariff share of US manufacturing imports experienced a differential increase of approximately 147% in Mexican exports following the Section 301 tariffs ($\hat{\beta} = 1.474$, $t = 9.14$), without a corresponding rise in Chinese exports into Mexico, which rules out transit deflection as the dominant mechanism. Heterogeneity analysis reveals two distinct mechanisms: demand-side substitution in low-capacity sectors, and input sourcing restructuring with a reduction in Chinese component dependence in high-capacity sectors. In particular, the article documents the emergence of hybrid routes in which Mexico's export expansion to the United States coexists with the persistent integration of Chinese inputs, suggesting a partial reconfiguration of regional GVCs. We examine the productive and technological dependence on China and discuss the implications of these findings for Mexican industrial policy.

Keywords: Nearshoring; Global Value Chains; Hybrid routes; U.S.-China trade war; Mexican manufacturing; industrial policy.

JEL Classification: F13, F14, O19

Resumen

Este artículo analiza la reconfiguración reciente de las Cadenas Globales de Valor (CGV) entre México, China y Estados Unidos en el contexto del T-MEC, del proceso de nearshoring y de las tensiones comerciales globales asociadas al denominado segundo shock de China. Utilizando datos bilaterales de comercio a nivel de producto (HS6) provenientes de UN Comtrade entre enero de 2016 y noviembre de 2024, identificamos cambios en los patrones de competencia directa entre México y China en el mercado estadounidense, así como en la dependencia de la manufactura mexicana respecto de insumos importados de China. Mediante un diseño de diferencias en diferencias con efectos fijos por producto y año-mes, estimamos que los sectores donde China concentraba una mayor participación previa en las importaciones estadounidenses registraron un crecimiento diferencial de aproximadamente 147% en las exportaciones mexicanas tras los aranceles de la Sección 301

*Facultad de Economía, Universidad Nacional Autónoma de México (UNAM). Email: vladimirmarquezst@comunidad.unam.mx. ORCID: 0009-0001-1187-6580.

**Facultad de Economía, Universidad Nacional Autónoma de México (UNAM). Email: seykasandoval@comunidad.unam.mx. ORCID: 0000-0003-1552-9823.

($\hat{\beta} = 1.474$, $t = 9.14$), sin que ello se acompañara de un incremento en las exportaciones chinas hacia México, lo que descarta la hipótesis de deflexión de comercio. El análisis de heterogeneidad revela dos mecanismos distintos: en sectores de baja capacidad productiva previa, se observa sustitución por el lado de la demanda; en sectores de alta capacidad, una reestructuración de insumos, con una reducción de la dependencia de componentes chinos. En particular, el artículo documenta la emergencia de rutas híbridas en las que la expansión exportadora de México hacia Estados Unidos coexiste con una integración persistente de insumos chinos, lo que sugiere una reconfiguración parcial de las CGV regionales. Se enfatiza la dependencia productiva y tecnológica respecto de China y se discuten las implicaciones de estos resultados para la política industrial mexicana.

Palabras clave: Nearshoring; Cadenas Globales de Valor; Rutas híbridas; Guerra comercial EE. UU.–China; manufactura mexicana; política industrial.

Clasificación JEL: F13, F14, O19.

1. Introduction

In 2023, Mexico surpassed China to become once again the largest exporter of goods to the United States. Although both North American countries already maintained high levels of economic integration, the trade war between the United States and China, initiated in 2017 through tariffs, export controls, and a policy of strategic decoupling, has redirected U.S. production linkages toward North America (Oropeza García, 2024). This phenomenon, termed nearshoring, has been defined as the regionalization of Global Value Chains (GVCs), initiated by cost differences and increasingly incorporating objectives of industrial policy, economic security, and strategic control of productive and technological assets (Gereffi, 2025; **?**; Xing, 2018).

This transition has been marked by inconsistent negotiations and changes in the strategic objectives of successive US administrations (Freund et al., 2025). In July 2018, during Donald Trump's first term, the US government lifted the tariff exemption on Mexican products established under the North American Free Trade Agreement (NAFTA). In response, Mexico implemented retaliatory measures against various US products, affecting bilateral trade estimated at over three trillion dollars annually. In February 2018, the United States initiated a new trade war against Mexico, imposing tariffs of 25 % on all imports from Mexico and Canada, except for oil and energy, which would be subject to a 10 % tariff. In 2025, the United States' average effective tariff rate increased from 2.3 percent in 2024 to 13.1 percent in June 2025 (Reinhart, 2025).

As part of this process, the United States Trade Representative (USTR) has systematically resorted to unilateral trade policy instruments, particularly Sections 201 and 301 of the Trade Act of 1974, to impose tariffs and other restrictive measures. These provisions authorize President Donald Trump to take .ªll appropriate measures,ïncluding trade retaliation, to force the elimination of practices considered discriminatory by foreign governments (Ruiz Estrada and Koutronas, 2025). The contemporary use of these tools has been described as part of a US industrial policy aimed at reconfiguring production chains, protecting economic sectors, and preserving technological advantages amid geopolitical competition (Ju et al., 2024). China, for its part, has deployed an explicit state framework of industrial policy that directs subsidies, preferential financing, and tax incentives to sectors such as semiconductors, robotics, biotechnology, electric vehicles, and aerospace, articulating productive, technological, and geopolitical objectives in its participation in GVCs (Autor et al., 2021; Kehoe and Xu, 2025).

Although this combination of industrial and trade policies, technological changes, and geopolitical competition has affected the Mexican manufacturing industry, the hybrid and multinational nature of GVCs makes it difficult to trace the effects of tariffs and trade wars on Mexican manufacturing. Mexico has lacked a comparable industrial strategy, operating as a functional space that both the United States and China employ to adjust their production and technological strategies. Mexican economic policy has prioritized FDI attraction and export-oriented manufacturing over the deliberate development of domestic productive and technological capabilities (Gereffi, 2025).

To evaluate Mexico's changing role within GVCs, it is necessary to understand how the dynamics of competition and dependence combine simultaneously along trade routes. The objective of this article is to analyze the reconfiguration of competition and dependence routes between Mexico and China with respect to exports of goods to the United States. In particular, we ask: in which sectors do Mexico and China compete most directly for the U.S. market? Which Mexican export sectors are most dependent on intermediate inputs from China to supply that market, and how are these hybrid routes configured? What recent changes have occurred in these routes as a result of the United States–Mexico–Canada Agreement (USMCA), nearshoring, or global trade tensions?

The article is organized as follows. Section 2 situates Mexico's participation in GVCs within the framework of the US-China trade dispute and the USMCA, reviewing the structural conditions that shape its productive and technological insertion. Section 3 describes the data and methodological strategy, including the construction of competition, dependence, and hybrid route indicators from bilateral product-level trade data, and the difference-in-differences design used to estimate the causal effect of Section 301 tariffs on Mexican and Chinese export flows. Section 4 presents the empirical results: we document the coexistence of direct competition between Mexico and China in the US market, estimate the differential expansion of Mexican exports in tariff-exposed sectors, and we characterize the persistence of Chinese input dependence across high- and low-capacity manufacturing sectors. Section 5 discusses the implications of these findings for trade and industrial policy, and examines the limitations of Mexico's current insertion strategy.

2. Background

Mexico's participation in GVCs has been a fixed element of the country's development strategy since the 1990s (Sandoval Cabrera, 2012). Trade liberalization was conceived as a strategy to attract foreign direct investment (FDI), promote competitiveness, and diversify exports through integration with the North American market. Since the early twenty-first century, China's growing integration into world trade has shifted the axis of GVCs toward East Asia, reducing the relative weight of North American manufacturing and configuring a new interregional competition (Garrido, 2022).

The current situation of Mexican manufacturing can be understood as the result of trade policy decisions at the regional level and adjustments induced by structural transformations in the world economy. *Nearshoring*, understood as the relocation of productive processes toward countries close to the destination market, has been conceived as a space for promoting economic development, in which the government articulates industrial and trade policy objectives, and where productive linkages favor regionalization within the USMCA framework (Hernández and Benítez, 2023). The *National Development Plan 2025–2030*, whose strategy centres on expanding infrastructure and regional integration, contains explicitly pro-industrial-policy and import-substitution language:

> "Noteworthy is the formation of Plan México, an initiative that envisages collaboration between the Government of Mexico and the private sector, whose objective is to foster equitable and sustainable long-term economic development, based on the use of our domestic market for our own industry, thereby reducing unnecessary imports. This Plan seeks to promote private investment, generate formal and quality employment, increase national content in productive processes, simplify administrative procedures, and combat poverty and inequality" (Gobierno de México, 2025, p. 17).

Public investment will continue "to play a central role in economic development, oriented toward strategic projects that promote regional industrialization and the strengthening of national content". The relocation of investments is conceived as .ªn opportunity to consolidate the country's productive and technological sovereignty", with emphasis placed on preserving fiscal orthodoxy around the autonomy of the Bank of Mexico, balanced budgets, and the stability of public debt relative to GDP (Gobierno de México, 2025, pp. 17–18).

While the Plan presents public investment as a central axis of the development strategy and of technological capacity-building, the evidence of its execution remains limited. Sectoral evidence suggests that Mexican firms continue to participate in the least profitable segments of the value chain, where technological spillovers and productive linkages are scarce (Sandoval Cabrera, 2019; Basave Kunhardt et al., 2023, 2024; Sandoval and Ortega, 2025). On the other hand, facing trade tensions between Mexico and the United States, multinational firms operating in Mexico have deferred investments and, in some cases, are considering relocating their factories abroad (Ando et al., 2024), although such displacement is limited for already-established investments (Shih, 2020; **?**).

In this context, Mexico's trade bloc with the United States has contributed to consolidating an institutional framework that structurally limits trade integration beyond North America.[1] USMCA introduced rules of origin and labor content requirements for the Mexican manufacturing sector, and treaty clauses prohibit the signing of trade agreements with states considered "non-market" economies, which in practice limits broad institutional rapprochement with China (Gómez Tovar and Ruiz Nápoles, 2021).

This institutional lock-in also leaves Mexico structurally exposed to shifts in US trade policy Rodríguez-Clare et al. (2025) estimate, using a dynamic model of trade and reallocation with nominal wage rigidities, that US tariff increases and retaliatory measures caused significant real income losses both in Mexico and in other close US trading partners. Flach and Baur (2025) use a quantitative trade model to simulate the impact of Trump's 25 % tariffs on Mexico and Canada, and find that retaliatory measures will reduce Mexico's manufacturing value added by up to 13 %, reflecting long-run losses from tariff escalation, including efficiency losses, substitution effects, declining exports, and a possible recession.

For Mexico, China is a central supplier of final goods, intermediate inputs, and capital goods, while Mexican exports to China are relatively scarce. In 2024, Mexico imported approximately 130 billion dollars (USD) in goods from China, generating a significant and persistent trade

[1]Within the USMCA region, there is a growing flow of FDI, particularly in reinvestments, the consolidation of a dynamic automotive and auto-parts sector, and the development of other strategic sectors such as electronics, aerospace, and medical devices.

deficit. This imbalance can be interpreted as a growing technological dependence of the Mexican manufacturing sector on Chinese-sourced inputs(Pérez-Santillán and Arriaga Navarrete, 2024). Analyses on Mexican manufacturing document a high dependence on imported inputs, explained endogenously by limited investment, particularly in science, technology, and innovation (STI) (Basave Kunhardt et al., 2024), in labour force capacities (Basave Kunhardt et al., 2023), and institutional environment (Kehoe and Xu, 2025).

This situation can be partially observed through the "deterioration of the terms of trade" hypothesis[2]. The international value-added structure of trade is articulated by the design and control of strategic technological and intermediate goods, produced or controlled through ownership by firms located in the so-called *core* countries, while low-value-added manufacturing is located based on the network *leadership* design, in the non-core *(peripheral)* countries. (Durán Lima and Zaclicever, 2013; Garrido, 2022).

For Mexico, the export of medium- and high-technology manufacturing has been extensively documented, alongside a high dependence on imported inputs of greater technological complexity and a specialization in assembly activities. This insertion character is reflected in the low share of domestic value added in manufacturing exports and in the predominance of backward linkages within GVCs (Castillo and de Vries, 2018; Blyde, 2014). In this sense, the deterioration of the terms of trade manifests in the country's limited capacity to generate significantly greater value added within the framework of its insertion into the GVCs of the USMCA region.

Based on this diagnosis, *Plan México* (2025) sets out an import substitution strategy, particularly with respect to China[3], which seeks to increase productive capacities, employment, worker compensation, and domestic value added, driven by USMCA constraints and the second Trump administration tariff policy. Tariffs and trade policies more broadly can, under these conditions, operate as instruments for reconfiguring productive location and inducing substitution across GVCs (Utar et al., 2023; Semieniuk, 2022).

This perspective draws on the strategic trade literature, which moves beyond viewing tariffs as distortions and reconceptualizes them as instruments of industrial transformation. Rodrik (2011) argues that tariffs can be instruments of "policy space": for the design of industrial and technological policies consistent with countries' own productive structures and levels of development, even if this temporarily implies deviating from free international competition. This revisits the models of Krugman (1983) and Brander and Spencer (1984, 1985); Eaton and Grossman (1986), which, departing from perfect competition, argue that, since manufacturing is subject to increasing returns, trade barriers can be absorbed by rents, allowing domestic

[2]This concept derives from limited productive and export specialization, which establishes an asymmetry between the high income-elasticity of imports and the low income-elasticity of exports (Prebisch, 1949), exhibiting a tendency toward structural balance-of-payments disequilibrium and external vulnerability. The idea of "structural external disequilibrium" was the element that gave rise to the notion of the "import substitution industrialisation process" (Noyola Vázquez, 1957; Sunkel, 1958), where the disequilibrium "is grounded in a basic diversity of productive structures: specialization and heterogeneity characterize the peripheral structure, in contrast to the diversification and homogeneity of the centre" (Rodríguez, 2006, p. 57). This pattern of industrial development generates limited horizontal diversification, intersectoral complementarity, and vertical integration of productive sectors, which do not permit rapid diversification of the peripheral countries export supply, which tends to preserve its primary character for prolonged periods (Bárcena et al., 2022).

[3]Tariff policy, in the context of international trade tensions, can be interpreted as an industrial policy instrument oriented toward protecting or developing strategic sectors, incentivizing domestic production, and reconfiguring participation in global value chains; however, its effects are heterogeneous and depend on the productive and institutional context (Semieniuk, 2022; Qiu et al., 2019).

firms to gain scale, provided they manage to increase their productivity and the government eventually withdraws protection, avoiding rent capture by less efficient firms. Policies such as export subsidies or selective restrictions can, under these conditions, improve national welfare at the expense of other countries, in the same way that firms use investment or R&D to gain advantages in oligopolistic markets.

In this context, Plan México can be understood as Mexico's reaction to the second China shock, a policy initiative that seeks to implement trade and industrial policy actions as a mediator of US policies to combat trade surpluses that overflow traditional comparative advantage explanations and reflect, instead, a combination of industrial subsidies, directed credit to manufacturing firms, and financial mechanisms that prevent the appreciation of the Chinese currency and limit the absorption of imports Although uncertainty persists regarding the causes of these imbalances, their most visible manifestation is the massive expansion of Chinese exports (Autor et al., 2016, 2021). In response, the predominant reaction in recipient countries has been the imposition of tariffs on Chinese goods, a measure supported by both protectionist sectors and various political currents (Freund et al., 2024).

Pettis (2024) asserts that trade imbalances during the second China shock cannot be understood solely through comparative advantage, but are linked to macroeconomic configurations such as the repression of domestic consumption, the transfer of income toward sectors with increasing returns to scale, and the accumulation of external surpluses. Pettis describes the misalignments between savings, investment, and aggregate demand, as well as on the role of the Chinese financial system and state in generating export overcapacity. From this perspective, the "China shocks" influence the Mexican development trajectory by reinforcing patterns of subordinate insertion into GVCs, where growth in export volumes coexist with technological dependence and limited value-added capture (Oropeza García, 2024).

Plan México and the reactivation of industrial policy can be justified as aimed at a transformation of manufacturing activity consisting of a compositional shift that, due to the extraordinary growth of Chinese export-oriented activities participating in global production processes, widened the divergence with activities not participating in such processes, oriented mainly toward the domestic market and locally integrated, affecting their economic importance, while their degree of integration declined(Xing, 2018). Kehoe and Xu (2025) illustrate this contrast directly: China's upgrading along the value chain has been driven by technological sophistication, infrastructure investment, and R&D promotion, while Mexico's growth has remained constrained by its dependence on exporting the same products as part of the North American supply chain.

Mexico's integration into GVCs is considered a pathway for manufacturing development; however, this assumption overlooks the hierarchical relations of production networks. Lead firms actively segment production, retain control over core technologies, and assign lower-value tasks to peripheral suppliers (Gereffi et al., 2005; Sandoval Cabrera, 2019). Knowledge does not diffuse automatically; it is the object of dispute, negotiation, and often blockage. For countries integrated primarily as assembly centers, GVC participation can generate employment and export revenues without generating domestic technological capacities (Humphrey and Schmitz, 2002). This pattern describes with precision Mexico's insertion into the North American supply chain: sustained export growth concentrated in low-value-added segments, with persistent dependence on externally sourced inputs and technologies, a dependence that the nearshoring process reorganizes but does not structurally transform."

The two China shocks can be interpreted as a manifestation of structural development problems in the case under study: asymmetric productive specialization, technological dependence, and external constraints. The first shock reflected China's insertion as a low-cost manufacturing platform, which displaced industrial production in advanced and semi-peripheral economies such as Mexico, deepening processes of deindustrialization worldwide (Autor et al., 2021). The second China shock is characterized by the expansion of productive capacity in medium- and high-technology sectors, driven by active industrial policies, subsidies, and directed credit, which intensifies global competition and reconfigures value chains. As a result, China's development trajectory has shifted from *catching up* to *forging ahead*, positioning itself as one of the two central players in the world economy, competing for the surplus of the dynamic economic sectors associated with the electronics-computing sector and artificial intelligence.

3. Methods

3.1. Data

We draw bilateral trade data from UN Comtrade at the product level (HS6), restricted to manufacturing industries (HS chapters 28–96). Three monthly flows are constructed for each sector s: exports from Mexico to the United States ($\mathrm{MEX{\to}US}_{s,t}$), exports from China to the United States ($\mathrm{CHN{\to}US}_{s,t}$), and imports into Mexico from China ($\mathrm{CHN{\to}MEX}_{s,t}$). We define two time windows: a pre-shock period $\mathcal{T}_0$ (January 2016 – December 2017) and a post-shock period $\mathcal{T}_1$ (July 2018 – November 2024), with 2018 treated as a transition year. For any variable $Z_{s,t}$, the window average is

$$\overline{Z_s^{(\tau)}} = \frac{1}{|\mathcal{T}_\tau|} \sum_{t \in \mathcal{T}_\tau} Z_{s,t}, \qquad \tau \in \{0,1\}. \tag{1}$$

Figure 1 classifies each sector s according to the active presence of trade flows in the pre-tariff period, defined as a monthly average value exceeding \$1,000, into five trade routes. For each sector the log growth rates are computed as

$$gX_s = \log\left(\overline{\mathrm{MEX{\to}US}_s^{(1)}} + \varepsilon\right) - \log\left(\overline{\mathrm{MEX{\to}US}_s^{(0)}} + \varepsilon\right), \tag{2}$$

$$gM_s = \log\left(\overline{\mathrm{CHN{\to}US}_s^{(1)}} + \varepsilon\right) - \log\left(\overline{\mathrm{CHN{\to}US}_s^{(0)}} + \varepsilon\right), \tag{3}$$

where $\varepsilon > 0$ avoids taking the logarithm of zero. A sector is classified as a *hybrid route* if the expansion of Mexican exports to the US market coexists with a persistent or growing dependence on Chinese inputs:

$$Hybrid_s = \mathbf{1}(gX_s > 0 \wedge gM_s \geq 0). \tag{4}$$

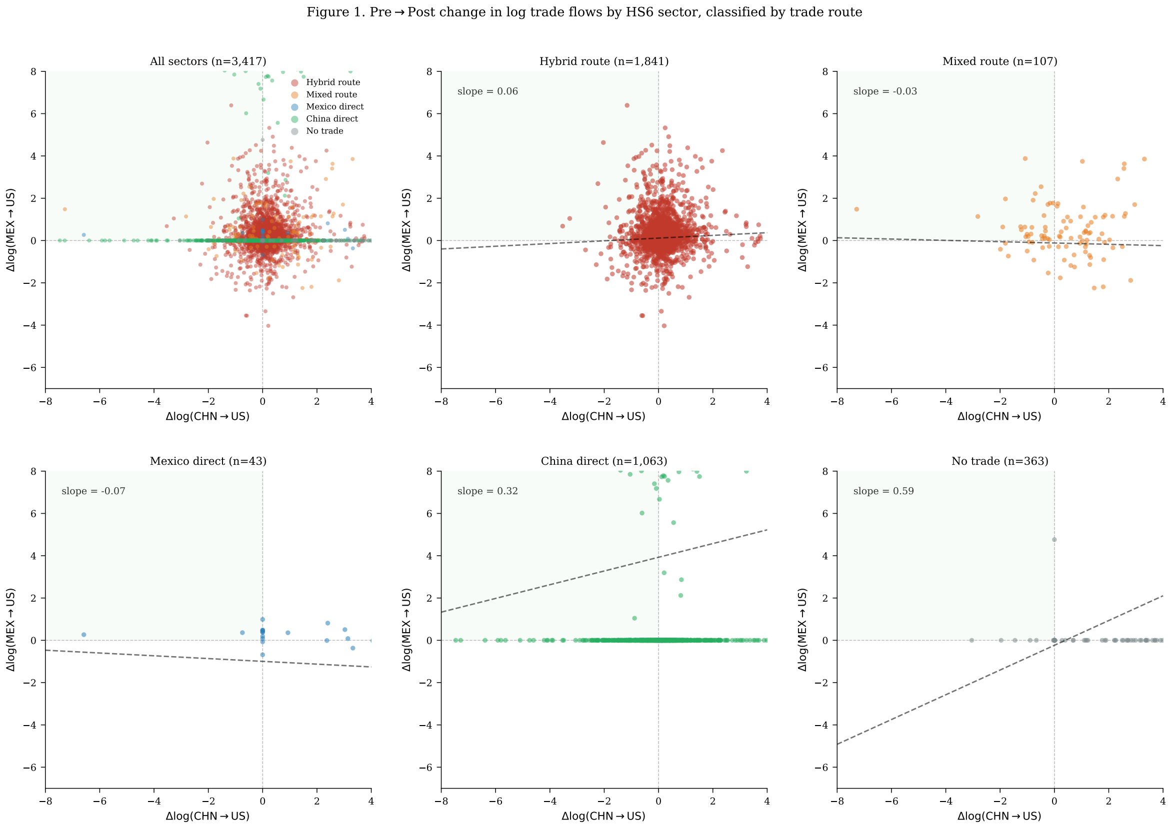

Figura 1: Pre/post shock change by HS6 category, disaggregated by trade route. Authors' own elaboration.

Figure 1 plots $\Delta\log(\text{CHN}{\rightarrow}\text{US})$ against $\Delta\log(\text{MEX}{\rightarrow}\text{US})$ for each HS6 manufacturing sector, disaggregated by pre-tariff trade route. In *hybrid route* sectors — where all three bilateral flows were active before the tariffs — the slope is near zero (0.08), indicating that changes in Chinese and Mexican export flows to the US were largely independent of one another, consistent with tariff-induced demand substitution rather than mechanical displacement. Sectors with only a Chinese pre-tariff US route (*China direct*, slope $= 0.33$) and sectors with no prior trade presence (*no trade*, slope $= 0.60$) show positive co-movement driven by common sectoral shocks and the mechanical amplification of small absolute changes in log space, respectively. The *Mexico direct* and *mixed route* panels show slopes close to zero, consistent with Mexican exporters responding to their own demand conditions independently of Chinese performance.

Table 1 presents summary statistics for the estimation sample. Our panel covers 1,937 HS6 manufacturing product codes across 107 months from January 2016 to November 2024, and yields 207,259 sector-months observations after restricting to codes present in all three trade flows. Pre-tariff monthly exports from Mexico to the US average \$34.92M at the HS6 level, compared to \$17.56M for Chinese exports to the US in the same sectors. China's mean pre-tariff share of combined US imports (the treatment variable) is 0.56, with substantial variation across sectors (SD $= 0.34$), ranging from near-zero in sectors where Mexico historically dominated to near-unity in sectors where Chinese goods faced little Mexican competition.

Figure 2 documents the aggregate trends of the three flows indexed to the 2016–2017 baseline. Total Mexican manufacturing exports to the US reached approximately 48 % above their pre-

Cuadro 1: Summary statistics

Panel A: Panel structure

	Value
Total observations	207,259
Unique HS6 product codes	1,937
HS2 sections	41
Months	107
Date range	Jan 2016 – Nov 2024
Post-tariff cutoff	Jul 2018 (Section 301 List 1)

Panel B: Trade flow values (USD millions, non-zero HS6-month observations)

Flow	Mean	SD	P25	Median	P75	n
Mexico → US	34.92	189.66	0.34	2.20	12.45	131,644
China → Mexico	2.28	11.24	0.06	0.30	1.16	172,151
China → US	17.56	109.26	0.31	1.83	8.55	192,989

Panel C: Treatment variable — China's pre-tariff US import share

	Mean	SD	P25	Median	P75	n
TreatShare_s	0.559	0.337	0.236	0.599	0.892	207,259

Panel D: Monthly aggregate exports, pre- and post-tariff (USD billions)

Period	Mexico → US	China → US	China → Mexico
Pre-tariff (2016–2017)	37.78	28.88	2.39
Post-tariff (Jul 2018–Nov 2024)	44.48	32.64	4.14
Change (%)	+17.6	+13.0	+73.2

Panel E: Productive capacity split

	HS6 codes
High capacity (pre-tariff MEX→US $\geq$ \$0.82M/month)	969
Low capacity (pre-tariff MEX→US $<$ \$0.82M/month)	968

Notes: Panel A restricts to HS6 codes present in all three bilateral trade flows. Panel B reports USD millions for non-zero HS6-month observations; structural non-trading zeros are retained in the regression. Panel C defines treatment as China's share of combined China+Mexico manufacturing imports to the US, averaged over 2016–2017. Panel D reports simple monthly averages across all panel HS6 codes. Panel E splits at the median pre-tariff Mexico→US export level (\$0.82M/month). All values from UN Comtrade; manufacturing defined as HS2 sections 28–30, 32–34, 38–40, 54–64, 72–80, and 84–96. December months are included in Panels B–D but excluded from regressions to remove Comtrade reporting issues.

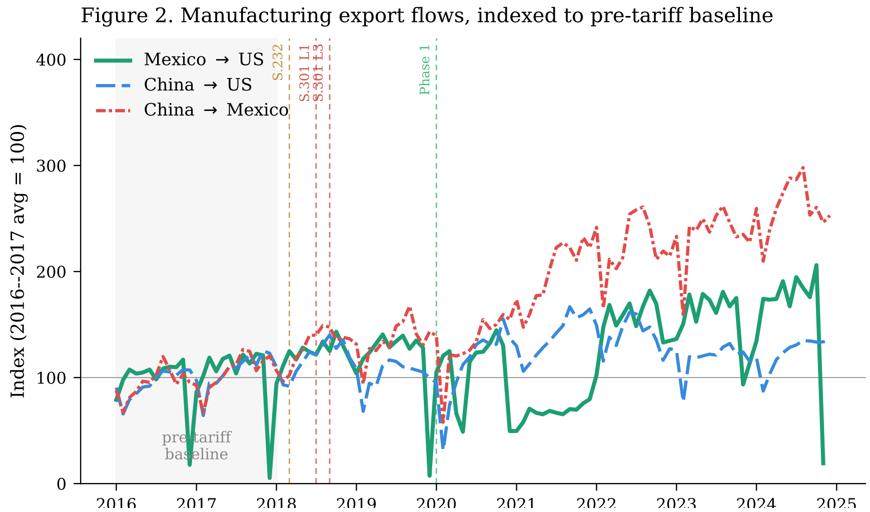

Figura 2: Manufacturing export flows, indexed to pre-tariff baseline

Notes: Monthly FOB values (USD) for manufacturing HS6 product codes across three bilateral trade flows: Mexico to the US (solid), China to the US (dashed), and China to Mexico (dash-dot). Index equals 100 at the 2016–2017 average. Vertical markers denote key policy events: Section 232 steel and aluminum tariffs (Mar 2018), Section 301 List 1 (Jul 2018), Section 301 List 3 (Sep 2018), and the US–China Phase One deal (Jan 2020). Shaded region is the pre-tariff baseline period. Source: UN Comtrade.

tariff level by 2024, while Chinese exports to the US in the same sectors were only 3 % above baseline. Chinese exports to Mexico, our deflection diagnostic grew substantially, reaching 116 % above baseline, but the composition, concentrated in intermediates rather than finished goods, indicates supply chain deepening rather than transit deflection. These descriptive patterns motivate the identification strategy.

3.2. Pre-trends validation

A credible difference-in-differences design requires that sectors with high and low Chinese pre-tariff US market share were on parallel trajectories before the tariff shock. Figure 3 presents the event study coefficients from estimating equation (5) over a 24-month pre-period and 30-month post-period window, with December months excluded to remove Comtrade reporting issues. Formally, we estimate:

$$\log(1+\text{MEX-US}_{s,t}) = \alpha_s + \alpha_t + \sum_{k\neq -1} \beta_k \left(\text{TreatShare}_s \times \mathbf{1}[t=k]\right) + \varepsilon_{s,t} \tag{5}$$

where all coefficients are normalized to zero at the pre-period mean. The pre-period coefficients fluctuate around zero without trend (pre-period mean $= 0.00$, SD $= 0.27$), providing no evidence of differential pre-trends between high- and low-exposure sectors. The post-period coefficients are initially close to zero and begin to rise from approximately month +15 onward, indicating gradual production relocation rather than an immediate mechanical response.

The corresponding event study for the deflection outcome (Figure 4) shows a flat pre-period followed by a persistent negative shift beginning around month +8 and deepening through 2022–2024. This pattern rules out immediate transit deflection, which would predict a positive post-period jump, and suggests supply chain restructuring away from Chinese inputs, as

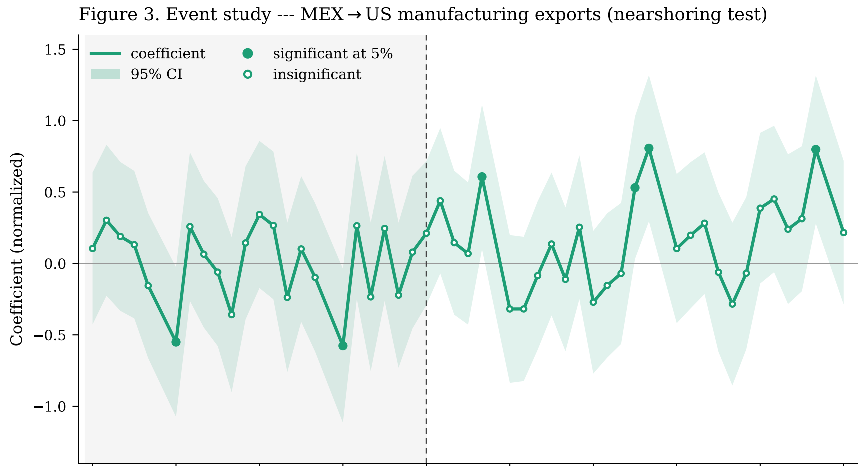

Figura 3: Event study — MEX→US manufacturing exports (nearshoring test)

Notes: Coefficients from interacting the pre-tariff Chinese US import share with period dummies, normalized so the pre-period mean equals zero. Two-way fixed effects (HS6 + year-month). Standard errors are clustered at the HS2 level (41 clusters). December months are excluded to remove Comtrade reporting issues. Filled circles denote months where the 95% confidence interval excludes zero. The shaded region is the pre-tariff period. $n = 191{,}763$.

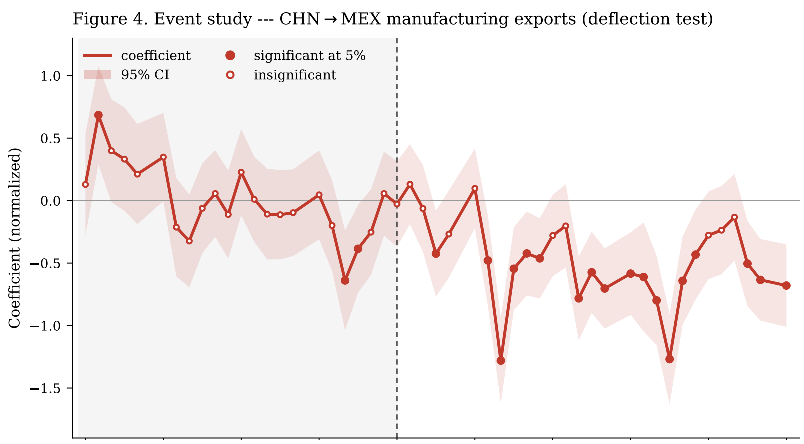

Figura 4: Event study — CHN→MEX manufacturing exports (deflection test)

Notes: Same specification as Figure 3. The outcome variable is Chinese exports into Mexico at the HS6-month level. The statistically significant negative shift beginning at approximately month +8 rules out transit deflection and indicates supply chain restructuring away from Chinese inputs as Mexican producers expand US-bound production.

Mexican production expands.

4. Results

Table 2 presents the main regression results from estimating the two-equation system. All specifications include HS6 product fixed effects and year-month fixed effects, absorbing time-invariant sector characteristics and all aggregate shocks common to manufacturing exports in a given month. Standard errors are clustered at the HS2 section level (41 clusters). Formally, we estimate:

Cuadro 2: Main results: effect of Chinese tariff exposure on Mexican and Chinese manufacturing trade flows

	(1) MEX→US (nearshoring test)	(2) CHN→MEX (deflection test)
$\text{TreatShare}_s \times \text{Post}_t$	1.474^{***}	-0.279
	(0.161)	(0.183)
HS6 fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Observations	247,170	247,170
HS6 codes	1,937	1,937
HS2 clusters	41	41

Notes: OLS estimates with two-way fixed effects (HS6 product and year-month), absorbed via iterative demeaning. Outcome in column (1) is $\log(1+\text{MEX} \to \text{US}_{s,t})$ and in column (2) is $\log(1+\text{CHN} \to \text{MEX}_{s,t})$. TreatShare_s is China's share of combined China+Mexico manufacturing imports to the US, averaged over 2016–2017. $\text{Post}_t = 1$ from July 2018 onward (Section 301 List 1). Standard errors in parentheses, clustered at the HS2 section level. Observation counts reflect non-zero trade flow observations after within-transformation. $^{***}p < 0.01$.

$$\log(1+\text{MEX-US}_{s,t}) = \alpha_s + \alpha_t + \beta\,(\text{TreatShare}_s \times \text{Post}_t) + \varepsilon_{s,t} \tag{6}$$

$$\log(1+\text{CHN-MEX}_{s,t}) = \alpha_s + \alpha_t + \gamma\,(\text{TreatShare}_s \times \text{Post}_t) + \varepsilon_{s,t} \tag{7}$$

where s indexes HS6 product codes and t indexes year-months. α_s are HS6 product fixed effects absorbing all time-invariant sector characteristics. α_t are year-month fixed effects absorbing aggregate shocks common to all sectors in a given month. TreatShare_s is China's share of combined China+Mexico manufacturing imports to the US averaged over the pre-tariff period (2016–2017), measuring the intensity of competitive displacement. $\text{Post}_t = 1$ from July 2018 onward (Section 301 List 1). The coefficient β in equation (6) tests whether Mexican exports to the US rose differentially in sectors most exposed to Chinese displacement (nearshoring). The coefficient γ in equation (7) tests whether Chinese exports into Mexico rose in the same sectors (deflection). Both equations are estimated separately via OLS with the two-way fixed effects absorbed through iterative demeaning.

Equation (6) — *Nearshoring test.* Column (1) reports the effect of Chinese tariff exposure on Mexican exports to the US. The coefficient $\hat{\beta}$ on $\text{TreatShare}_s \times \text{Post}_t$ is 1.474 (SE = 0.161, $t = 9.14$, $p < 0.01$). Sectors where China held a one-unit higher pre-tariff share of US manufacturing imports saw approximately 147 % more growth in log Mexican exports after July 2018, relative to sectors where Mexico already dominated. This estimate is robust across specifications.

Equation (7) — *Deflection test.* Column (2) reports the parallel equation for Chinese exports into Mexico. The coefficient $\hat{\gamma}$ is -0.279 (SE = 0.183, $t = -1.52$), which is negative but statistically insignificant. In sectors most displaced from the US market, Chinese exports into Mexico did not rise, the opposite of what transit deflection would predict. This constitutes the paper's core falsification result.

Taken together, the two equations point to a nearshoring mechanism: sectors where Chinese goods lost US market access saw more Mexican exports to the US and simultaneously showed no increase in Chinese goods flowing into Mexico. The joint pattern is inconsistent with the null of no adjustment and the alternative of transit deflection.

4.1. Heterogeneity by productive capacity

The aggregate estimates mask heterogeneity in the adjustment mechanism. If nearshoring reflects production relocation to Mexico, the response should differ by Mexico's pre-existing productive capacity in each sector, both because established Mexican manufacturers can more rapidly absorb displaced demand, and because capacity is a proxy for supply chain readiness. Table 3 splits the sample at the median pre-tariff MEX→US export level ($0.82M per month) as a measure of Mexico's pre-existing productive capacity. The results reveal two distinct patterns

In *low-capacity sectors*, those where Mexico had minimal pre-tariff export presence, the nearshoring coefficient is large and significant ($\beta = 1.399$, $t = 5.56^{***}$), while the deflection coefficient is negative but insignificant ($\beta = -0.234$, $t = -1.06$). Mexican exports surged in precisely the sectors where China was most displaced, without a corresponding rise in Chinese goods entering Mexico. This is consistent with new entry, demand capture, or rapid capacity expansion by Mexican manufacturers responding to opportunities opened by the tariffs.

In *high-capacity sectors*, those where Mexico already had established export positions, the nearshoring coefficient is positive but statistically insignificant ($\beta = 0.388$, $t = 1.46$), while the deflection coefficient is large and highly significant ($\beta = -1.545$, $t = -4.91^{***}$). Established Mexican manufacturers show no additional export surge, but the fall in Chinese imports into Mexico suggests they actively restructured their input sourcing, reducing reliance on Chinese components as they deepened domestic or non-Chinese supply chains.

We interpret the following two mechanisms: Low-capacity sectors experienced demand-side substitution: Mexican producers captured the markets vacated by Chinese exporters. High-capacity sectors experienced supply-side restructuring: established Mexican manufacturers reorganized their input networks independent of any export volume response. Both are consistent with integration deepening rather than trade deflection.

Cuadro 3: Heterogeneity by Mexican pre-tariff productive capacity

	High capacity		Low capacity	
	(1) MEX→US	(2) CHN→MEX	(3) MEX→US	(4) CHN→MEX
TreatShare$_s \times$ Post$_t$	0.388	-1.545^{***}	1.399^{***}	-0.234
	(0.266)	(0.315)	(0.252)	(0.221)
HS6 fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Observations	104,432	104,432	142,738	142,738
HS6 codes	969	969	968	968
HS2 clusters	38	38	41	41

Notes: Same specification as Table 2, estimated separately on high- and low-capacity subsamples. High (low) capacity denotes HS6 codes where Mexico's pre-tariff monthly exports to the US were above (below) the sample median of \$0.82M/month, used as a proxy for Mexico's pre-existing productive capacity in that sector. Standard errors in parentheses, clustered at the HS2 section level. In high-capacity sectors, Mexican export volumes do not surge (insignificant) but Chinese inputs into Mexico fall significantly. In low-capacity sectors, Mexican exports rise strongly without a corresponding increase in Chinese imports into Mexico. $^{***}p < 0.01$.

4.2. Robustness

Table 4 assesses the sensitivity of the main estimates to specification choices.

Excluding December months. Comtrade reporting patterns produce systematically low or missing trade values in December for many HS6 codes. Dropping all December observations, 15,496 sector-months, approximately 7.5 % of the sample, leaves the nearshoring coefficient essentially unchanged at 0.660 ($t = 3.65$, moving from the baseline of 1.474). The deflection coefficient strengthens to -1.076 ($t = -5.94$), remaining highly significant. The stability of the nearshoring result and the strengthening of the deflection result confirm that December reporting gaps do not drive the main findings.

Alternative treatment variable. As an alternative to the pre-tariff share measure, we construct a treatment variable based on the realized decline in China's US import share from 2017 to 2019, capturing competitive displacement. The nearshoring coefficient rises to 1.590 (SE = 0.332, $t = 4.79^{***}$), consistent with the realized displacement measure concentrating identification on sectors where the tariff shock was binding. The deflection coefficient is -0.233 but insignificant ($t = -0.49$), confirming the absence of systematic transit rerouting.

Dropping automotive (HS87). Vehicles and parts represent Mexico's largest manufacturing export category and are subject to USMCA-specific rules of origin that may confound the tariff identification. Dropping HS87 entirely removes 4,554 observations (2.4 % of the sample). The nearshoring coefficient is 0.701 (SE = 0.185, $t = 3.79^{***}$) and the deflection coefficient is -1.014 (SE = 0.180, $t = -5.65^{***}$), both virtually identical to the no-December baseline. Automotive sector dynamics do not drive the core results.

Cuadro 4: Robustness checks

	Baseline		(1) Drop December		(2) Alt. treatment		(3) Drop HS87	
	MEX→US	CHN→MEX	MEX→US	CHN→MEX	MEX→US	CHN→MEX	MEX→US	CHN→MEX
$\text{TreatShare}_s \times \text{Post}_t$	1.474***	−0.279	0.660***	−1.076***	1.590***	−0.233	0.701***	−1.014***
	(0.161)	(0.183)	(0.181)	(0.181)	(0.332)	(0.477)	(0.185)	(0.180)
HS6 FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	247,170		191,763		165,924		187,209	
HS2 clusters	41		41		41		40	

Notes: All specifications use OLS with two-way fixed effects (HS6 product and year-month) and standard errors clustered at the HS2 section level. Outcome variables are $\log(1+\text{MEX} \rightarrow \text{US})$ and $\log(1+\text{CHN} \rightarrow \text{MEX})$. The baseline replicates Table 2. *(1) Drop December:* excludes December months to remove Comtrade reporting issues; the nearshoring estimate falls to 0.660 but remains highly significant. *(2) Alt. treatment:* replaces TreatShare_s with the realized decline in China's US import share from 2017 to 2019 (ΔShare_s), capturing actual competitive displacement; the nearshoring coefficient rises to 1.590. *(3) Drop HS87:* excludes vehicles and parts, subject to USMCA-specific rules of origin; results are virtually unchanged. Standard errors in parentheses. $^{***}p < 0.01$.

5. Discussion

The results establish three facts about the Mexico–China–US trade triangle in the Section 301 tariff episode. First, Mexican manufacturing exports to the US grew significantly faster in sectors where Chinese goods faced the largest US market displacement, a large effect ($\hat{\beta} = 1.474$), precisely estimated ($t = 9.14$), and robust to alternative treatment definitions and sample restrictions. Second, this increase in Chinese goods did not accompany expansion flowing into Mexico, ruling out transit deflection as the dominant mechanism in the aggregate. Third, the heterogeneity by productive capacity reveals that these facts conceal different processes operating in different parts of the manufacturing sector.

In the aggregate, our results suggest that Mexico benefited from the US-China trade decoupling. From a structural standpoint, the result is consistent with the Eaton–Kortum-type trade models in which a tariff-induced relative price change generates trade diversion toward third countries with a comparative advantage in the affected sectors (Eaton and Kortum, 2002). Mexico's geographic proximity, the USMCA preferential access, and the existing manufacturing base positioned it to absorb displaced demand rapidly relative to other candidate countries. The magnitude of the effect, a one-unit increase in China's pre-tariff share associated with a 147% differential increase in log Mexican exports, is large enough to suggest that tariff-induced demand shifts, rather than pure price effects, are the operating mechanisms.

In *low-capacity sectors*, where Mexico had low pre-tariff export presence, the nearshoring coefficient is large and significant ($\hat{\beta} = 1.399$, $t = 5.56$) while the deflection coefficient is insignificant, which suggests demand-side substitution: Mexican producers, or new entrants to the sector, captured market space from Chinese exporters without drawing substantially on Chinese inputs. In *high-capacity sectors*, where established Mexican manufacturers already held US export positions, the nearshoring coefficient is insignificant ($\hat{\beta} = 0.388$, $t = 1.46$) while the deflection coefficient turns large and negative ($\hat{\gamma} = -1.545$, $t = -4.91$). Established producers significantly reduced their dependence on Chinese inputs. This supply-side restructuring without an export surge can be understood as manufacturers diversifying in anticipation of further trade policy uncertainty, or reallocating existing capacity rather than expanding it.

These mechanisms have different implications for the sustainability and the welfare effects of nearshoring. The demand-side substitution in low-capacity sectors reflects capacity expansion, new investment, supplier relationships, and employment in Mexican manufacturing. If so, the nearshoring gains in these sectors may be durable even if US-China trade tensions ease. The supply chain restructuring in high-capacity sectors is more ambiguous: a reduction in Chinese input dependence could reflect diversification or fragmentation, depending on whether Chinese inputs were priced competitively or were used for reasons of technical specificity. To distinguish these interpretations, firm-level data on input prices and productivity is required, which lies beyond the scope of this study.

Among the limits of this interpretation, the treatment variable, China's pre-tariff US import share, captures exposure to potential displacement but not actual tariff incidence at the HS6 level, since Section 301 rates varied within manufacturing and some HS6 codes were exempted. The alternative treatment was realized from 2017 to 2019 share changes partially address this, providing a larger nearshoring estimate ($\hat{\beta} = 1.590$), consistent with the main specification. The analysis also cannot separately identify the extensive margin (new Mexican exporters entering displaced sectors) from the intensive margin (existing exporters increasing volumes), a distinction relevant for evaluating the aggregate productivity effects. Finally, the panel ends in

November 2024, before the full impact of potential second-term tariff escalation; the durability of the patterns documented here will depend partly on the three countries' policy trajectory.

Taken together, the evidence supports a reading of the 2018–2024 episode as real nearshoring, a restructuring of North American manufacturing supply chains in response to tariff-induced relative price changes, rather than a statistical artifact or transshipment. The finding that this restructuring operated through different mechanisms in established versus emerging manufacturing sectors has implications for industrial policy: that interventions aimed at accelerating nearshoring may be better targeted at the low-capacity sectors where demand substitution is already occurring organically, and where investment subsidies could amplify and consolidate capacity gains.

6. Conclusions

Overall, the dynamics of nearshoring have affected Mexico's integration routes into GVCs, creating opportunities but also revealing limitations associated with structural dependence (Hernández and Benítez, 2023). The two China shocks provide the structural breaks for these dynamics: the first displaced Mexican manufacturing by introducing China as a low-cost platform; the second, characterized by Chinese industrial upgrading into medium- and high-technology sectors, has intensified competitive pressure while reshaping the input structure of Mexican manufacturing. The coexistence of competition with a persistent dependence on Chinese inputs indicates that Mexico's export growth to the United States is sustained by external production rather than domestic capability accumulation.

Our classification of products into hybrid routes captures this tension. Hybrid route sectors, where Mexican export expansion to the United States coexists with a persistent or increasing dependence on Chinese inputs are the dominant pattern in high-capacity manufacturing. In these sectors, nearshoring has proceeded as a reorganization of the triangular relationship: Mexico absorbs displaced US demand while remaining embedded in Chinese-controlled input networks. In contrast, low-capacity sectors show a clearer substitution dynamic, suggesting that productive disconnection from China is more likely where Mexican manufacturers were already operating at the margin of the supply chain. This set of findings is consistent with a narrative of partial nearshoring or regional rearticulation of value chains, rather than complete productive transformation.

Our initial framing described *nearshoring* as a reassertion of hegemonic control over strategic assets and innovation rents, in which control over technology, standards, and intangible assets remains concentrated in US and Chinese firms, reproducing a pattern of asymmetric integration. In this sense, Mexico's insertion into GVCs continues to be predominantly *passive*,[4] oriented toward the productive and geopolitical objectives of the United States, with Mexican industrial and trade policy operating, *de facto*, as an appendage of US industrial strategy.

This integration is reinforced by the USMCA guidelines, which codify productive integration with the United States and a reduced scope for Mexican trade diversification, and the erratic character of recent negotiations between Mexico and the United States. These external

[4]By passive integration we mean Mexico's participation, whether through its territory, foreign direct investment, or domestic firms, in non-strategic links of global value chains; that is, in those segments that do not constitute the origin of extraordinary or relative surpluses and that therefore do not generate domestic productive capacities capable of translating into sustained increases in labour productivity or greater domestic value added.

constraints are compounded by internal limitations that Mexico's "new industrial policy" fails to address. Mexico maintains a high degree of productive heterogeneity, with a business environment dominated by small and medium enterprises with a history of low productivity. However, the focus of Plan México is on strategic sectors linked to exports to the United States, and lacks a clear policy for national productive integration, beyond an import substitution narrative that has consisted of a tariff-based trade policy within a low-public-investment, fiscally conservative framework.

We conclude that export growth and Mexico's position as the United States' main trading partner do not substantially modify the dependent manufacturing structure, but mainly reorganize it. The persistence of hybrid routes under conditions of the second China shock suggests that the reorganization is asymmetric: Mexico captures the export volume while China retains the input rents. Nearshoring risks consolidating a pattern of subordinate specialization, in which Mexico finds itself caught between the industrial strategies of its two dominant partners and lacks the capacity to modify the conditions of its own integration. The absence of longer-term policy commitments suggests that industrial policy remains at a superficial level, responding primarily to short-term reorganizations of US production rather than translating into a strategy for building technological and labor capabilities. The results suggest that a window of opportunity exists, but it is narrow and closing. Capitalizing on it requires more than passive FDI attraction: Mexico needs an active industrial policy that invests in critical infrastructure, identifies strategic niches within GVCs, supports domestic firms in acquiring productive and technological capabilities, and implementing a human capital strategy capable of generating the skilled workforce, these transitions demand (Gereffi, 2025). This means developing the bureaucratic, technological, and private-sector capacities necessary to advance Mexico's own objectives through international and public-private partnerships, not merely to serve as a functional space for other countries' industrial strategies.

Declaration of generative AI use

During the preparation of this work, the authors used Claude (Anthropic) for coding and data processing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Author contributions

Vladimir Márquez Stone: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft. **Seyka Sandoval:** Conceptualization, Supervision, Writing – review and editing.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The bilateral trade data used in this study are publicly available from UN Comtrade Plus (https://comtradeplus.un.org/). Replication code is available at [https://github.com/vladimir-marquez-stone/hybridroutes-article].

References

Ando, M., Hayakawa, K., Kimura, F., 2024. Supply chain decoupling: Geopolitical debates and economic dynamism in East Asia. Asian Econ. Policy Rev. 19, 62–79. doi:10.1111/aepr.12434.

Autor, D.H., Dorn, D., Hanson, G.H., 2016. The China shock: Learning from labor-market adjustment to large changes in trade. Annu. Rev. Econ. 8, 205–240. doi:10.1146/annurev-economics-080315-015041.

Autor, D.H., Dorn, D., Hanson, G.H., 2021. On the persistence of the China shock. Working Paper w29401. National Bureau of Economic Research. doi:10.3386/w29401.

Bárcena, A., Bielschowsky, R., Torres, M., 2022. El pensamiento de la CEPAL (2009–2018): hacia una estrategia neoestructuralista de desarrollo basada en un enfoque de derechos. El Trimest. Econ. 89, 73–109. doi:10.20430/ete.v89i353.1416.

Basave Kunhardt, J., Montiel, P., Sandoval, S., 2024. Implementación de la Política Industrial Digital en México: desafíos y estrategias, in: Oropeza García, A. (Ed.), Hacia una nueva visión del desarrollo industrial digital en México. INADI / Consejo Coordinador Empresarial.

Basave Kunhardt, J., Montiel Martínez, P., Sandoval, S., 2023. De la atracción de la IED a la incorporación de valor agregado nacional, in: Basave, J. (Ed.), Balance de la política económica en México. UNAM, Instituto de Investigaciones Económicas.

Blyde, J., 2014. Mexico's participation in global supply chains: The challenge of adding Mexican value. Appl. Econ. Lett. 21, 501–504. doi:10.1080/13504851.2013.879274.

Brander, J.A., Spencer, B.J., 1984. Tariff protection and imperfect competition, in: Kierzkowski, H. (Ed.), Monopolistic Competition and Product Differentiation and International Trade. Oxford University Press, Oxford, pp. 194–206.

Brander, J.A., Spencer, B.J., 1985. Export subsidies and international market share rivalry. J. Int. Econ. 18, 83–100. doi:10.1016/0022-1996(85)90006-6.

Castillo, J., de Vries, G.J., 2018. The domestic content of Mexico's maquiladora exports: A long-run perspective. J. Int. Trade Econ. Dev. 27, 200–219. doi:10.1080/09638199.2017.1371853.

Durán Lima, J.E., Zaclicever, D., 2013. La América Latina y el Caribe en las cadenas internacionales de valor. CEPAL Ser. Comer. Int. , 1–67.

Eaton, J., Grossman, G.M., 1986. Optimal trade and industrial policy under oligopoly. Q. J. Econ. 101, 383–406. doi:10.2307/1891121.

Eaton, J., Kortum, S., 2002. Technology, geography, and trade. Econometrica 70, 1741–1779. doi:10.1111/1468-0262.00352.

Flach, L., Baur, A., 2025. The effect of Trump tariffs on Mexico and Canada. Policy Brief 70. EconPol.

Freund, C., Mattoo, A., Mulabdic, A., Ruta, M., 2024. Is US trade policy reshaping global supply chains? J. Int. Econ. 152, 104011. doi:10.1016/j.jinteco.2024.104011.

Freund, C., Mattoo, A., Mulabdic, A., Ruta, M., Wilczynska, K., 2025. Enduring dependence: De minimis, transshipment, and US-China trade. State Glob. .

Garrido, C., 2022. México en la fábrica de América del Norte y el nearshoring. Technical Report 48056. CEPAL.

Gereffi, G., 2025. Oportunidades y dificultades del nearshoring en México. Foro Int. 65, 925–972. doi:10.24201/fi.v65i4.2986.

Gereffi, G., Humphrey, J., Sturgeon, T., 2005. The governance of global value chains. Rev. Int. Polit. Econ. 12, 78–104. doi:10.1080/09692290500049805.

Gobierno de México, 2025. Plan Nacional de Desarrollo 2025–2030. Gobierno de México, México.

Gómez Tovar, R., Ruiz Nápoles, P., 2021. Efectos potenciales de los cambios en el T-MEC respecto al TLCAN sobre la economía mexicana. Norteamérica 16, 347–373. doi:10.22201/cisan.24487228e.2021.2.491.

Hernández, A.F., Benítez, E.B., 2023. Potencial del nearshoring para el desarrollo económico de México. Región Cient. 2, 2023105. doi:10.58763/rc2023105.

Humphrey, J., Schmitz, H., 2002. How does insertion in global value chains affect upgrading in industrial clusters? Reg. Stud. 36, 1017–1027. doi:10.1080/0034340022000022198.

Ju, J., Ma, H., Wang, Z., Zhu, X., 2024. Trade wars and industrial policy competitions: Understanding the US-China economic conflicts. J. Monet. Econ. 141, 42–58. doi:10.1016/j.jmoneco.2023.11.003.

Kehoe, T.J., Xu, X., 2025. Openness and growth: A comparison of the experiences of China and Mexico. Working Paper w34181. National Bureau of Economic Research. doi:10.3386/w34181.

Krugman, P.R., 1983. Targeted industrial policies: Theory and evidence. Ind. Change Public Policy , 123–155.

Noyola Vázquez, J., 1957. Inflación y desarrollo económico en Chile y México. Panorama Econ. 11, 273–286.

Oropeza García, A., 2024. El nearshoring. resultado de una confrontación geopolítica, in: Oropeza García, A. (Ed.), Nearshoring: La oportunidad de un nuevo desarrollo económico y social para México. UNAM, Instituto de Investigaciones Jurídicas, México, pp. 18–39.

Pérez-Santillán, L., Arriaga Navarrete, R., 2024. Integración productiva y comercial de México en las cadenas globales de valor. El Trimest. Econ. 91, 663–703. doi:10.20430/ete.v91i363.1877.

Pettis, M., 2024. China's coming economic adjustment, in: Critical Issues in Contemporary China. Routledge, London, pp. 48–60.

Prebisch, R., 1949. El desarrollo económico de la América Latina y algunos de sus principales problemas. El Trimest. Econ. 16, 347–431.

Qiu, L.D., Zhan, C., Wei, X., 2019. An analysis of the China–US trade war through the lens of the trade literature. Econ. Polit. Stud. 7, 148–168. doi:10.1080/20954816.2019.1595329.

Reinhart, A., 2025. US focus: Early check-in on realized tariff rates. Research Note. J.P. Morgan Economic Research.

Rodríguez, O., 2006. El estructuralismo latinoamericano. Siglo XXI, México.

Rodríguez-Clare, A., Ulate, M., Vásquez, J.P., 2025. The 2025 trade war: Dynamic impacts across US states and the global economy. Working Paper w33792. National Bureau of Economic Research. doi:10.3386/w33792.

Rodrik, D., 2011. The globalization paradox: Democracy and the future of the world economy. World Trade Rev. 10, 409–417. doi:10.1017/S1474745611000048.

Ruiz Estrada, M.A., Koutronas, E., 2025. International trade closeness: The impact of tariff policies on US–China trade relations. SSRN Available at SSRN 5167487. https://doi.org/10.2139/ssrn.5167487.

Sandoval, S., Ortega, A.Y.D., 2025. Capacidades locales y escalamiento: claves para una política industrial efectiva en México. Norteamérica 20, 5. doi:10.22201/cisan.24487228e.2025.2.629.

Sandoval Cabrera, S.V., 2012. Gobernabilidad y ascenso en la cadena de valor: una discusión conceptual. Anál. Econ. 27, 7–23.

Sandoval Cabrera, S.V., 2019. Upgrading y competencia: reflexiones para firmas y países en desarrollo. Econ. UNAM 16, 213–238. doi:10.22201/fe.24488143e.2019.48.484.

Semieniuk, G., 2022. Industrial policies and the competition for low-carbon manufacturing. Joule 6, 713–715. doi:10.1016/j.joule.2022.03.007.

Shih, W.C., 2020. Bringing manufacturing back to the U.S. is easier said than done. Harv. Bus. Rev. https://hbr.org/2020/04/bringing-manufacturing-back-to-the-u-s-is-easier-said-than-done (accessed 15 April 2020).

Sunkel, O., 1958. La inflación chilena: un enfoque heterodoxo. El Trimest. Econ. 25, 570–599.

Utar, H., Cebreros Zurita, A., Torres Ruiz, L.B., 2023. The US-China trade war and the relocation of global value chains to Mexico. Preprint. SSRN. doi:10.2139/ssrn.4568757.

Xing, Y., 2018. Global value chains and new thinking on trade and industrial policy, in: Xing, Y. (Ed.), New Humanism and Global Governance. World Scientific, Singapore, pp. 185–204.